

Data exploration and cleaning :

```
In [6]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import os

for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

/kaggle/input/aerofit-data/aerofit_treadmill.csv

```
In [7]: df = pd.read_csv("/kaggle/input/aerofit-data/aerofit_treadmill.csv")
df.head()
```

```
Out[7]:
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

```
In [8]: sns.set(style="whitegrid")
```

```
In [9]: df.shape
```

```
Out[9]: (180, 9)
```

```
In [10]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Product         180 non-null   object
1   Age             180 non-null   int64
2   Gender          180 non-null   object
3   Education       180 non-null   int64
4   MaritalStatus   180 non-null   object
5   Usage          180 non-null   int64
6   Fitness         180 non-null   int64
7   Income         180 non-null   int64
8   Miles          180 non-null   int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

```
In [11]: df.describe()
```

Out [11]:

	Age	Education	Usage	Fitness	Income	Miles
count	180.000000	180.000000	180.000000	180.000000	180.000000	180.000000
mean	28.788889	15.572222	3.455556	3.311111	53719.577778	103.194444
std	6.943498	1.617055	1.084797	0.958869	16506.684226	51.863636
min	18.000000	12.000000	2.000000	1.000000	29562.000000	21.000000
25%	24.000000	14.000000	3.000000	3.000000	44058.750000	66.000000
50%	26.000000	16.000000	3.000000	3.000000	50596.500000	94.000000
75%	33.000000	16.000000	4.000000	4.000000	58668.000000	114.750000
max	50.000000	21.000000	7.000000	5.000000	104581.000000	360.000000

MaritalStatus , Product and Gender are converted to "category" type

```
In [12]: categorical_cols = ["Product", "Gender", "MaritalStatus"]

for col in categorical_cols:
    df[col] = df[col].astype("category")

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Product         180 non-null    category
1   Age             180 non-null    int64
2   Gender          180 non-null    category
3   Education       180 non-null    int64
4   MaritalStatus   180 non-null    category
5   Usage           180 non-null    int64
6   Fitness         180 non-null    int64
7   Income          180 non-null    int64
8   Miles           180 non-null    int64
dtypes: category(3), int64(6)
memory usage: 9.5 KB
```

```
In [13]: df.describe(include="category")
```

Out [13]:

	Product	Gender	MaritalStatus
count	180	180	180
unique	3	2	2
top	KP281	Male	Partnered
freq	80	104	107

Summary

This dataset contains 180 observations with no missing values across all variables.

The average age is about 29 years, with most individuals falling between their mid-20s and early-30s. The youngest is 18, the oldest 50, so this skews young-to-mid adult rather than evenly spread across ages.

Education averages around 15.6 years, which roughly corresponds to some college or a bachelor's degree, with relatively low variability. Most values cluster tightly between 14 and 16 years, suggesting a fairly homogeneous educational background.

Usage has a mean of 3.46, with a range from 2 to 7. This implies moderate engagement on average, but with a noticeable tail of heavier users.

Fitness averages 3.31 on a scale from 1 to 5, centered strongly around 3–4. This suggests the population generally considers itself moderately fit, with fewer extreme low or high ratings.

Income averages about

53,700, but the high standard deviation tells an important story: incomes vary widely, from 44k and 59k, while the maximum exceeds 100k, indicating a right-skewed distribution.

Miles (likely distance-related usage or activity) has a mean of 103, but also shows substantial spread. Most users fall between 66 and 115 miles, yet the maximum of 360 miles suggests a small number of very high-activity individuals pulling the average upward.

In short: This is a young, moderately educated, moderately active population with fairly consistent fitness levels, wide income variation, and a small group of heavy users driving the upper extremes, especially in miles and usage.

```
In [14]: df.nunique()
```

```
Out[14]: Product      3
         Age          32
         Gender        2
         Education     8
         MaritalStatus  2
         Usage         6
         Fitness       5
         Income       62
         Miles        37
         dtype: int64
```

Analysis for Categorical Variables

Product Distribution

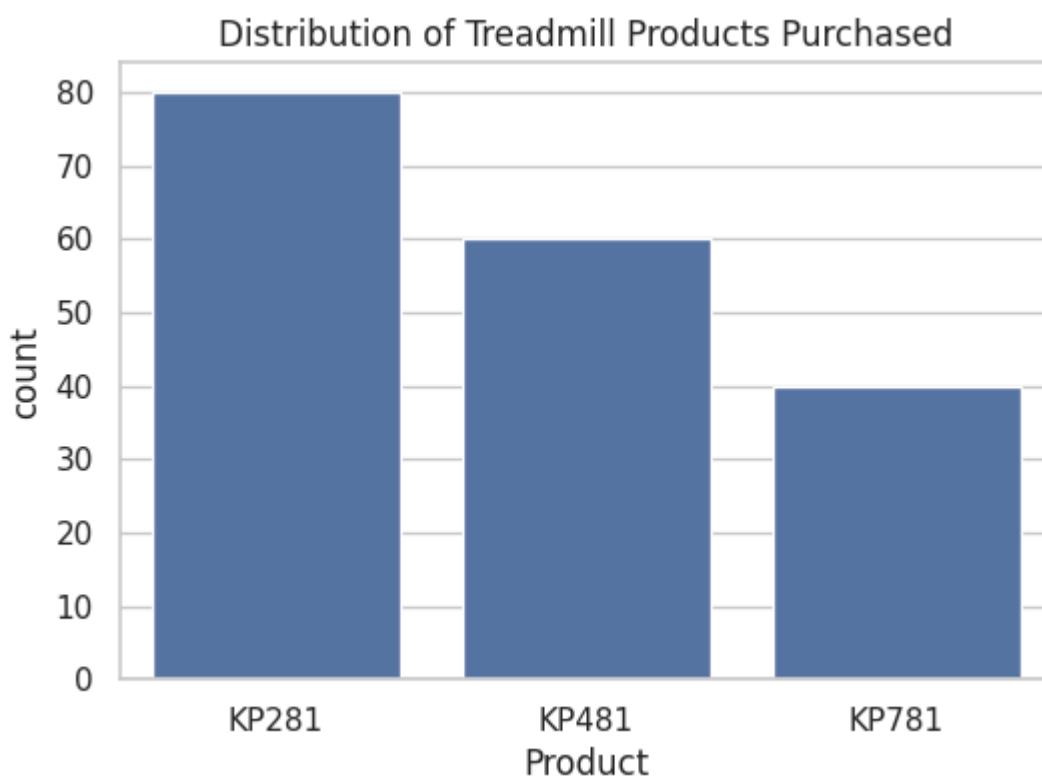
```
In [15]: df["Product"].value_counts()
```

```
Out[15]: Product
        KP281    80
        KP481    60
        KP781    40
        Name: count, dtype: int64
```

```
In [16]: df["Product"].value_counts(normalize=True) * 100
```

```
Out[16]: Product
        KP281    44.444444
        KP481    33.333333
        KP781    22.222222
        Name: proportion, dtype: float64
```

```
In [17]: plt.figure(figsize=(6,4))
sns.countplot(data=df, x="Product")
plt.title("Distribution of Treadmill Products Purchased")
plt.show()
```



Summary :

- ~45% customers bought KP281 → entry-level dominates.
- KP781 has the lowest share → premium product, niche audience

Gender Distribution

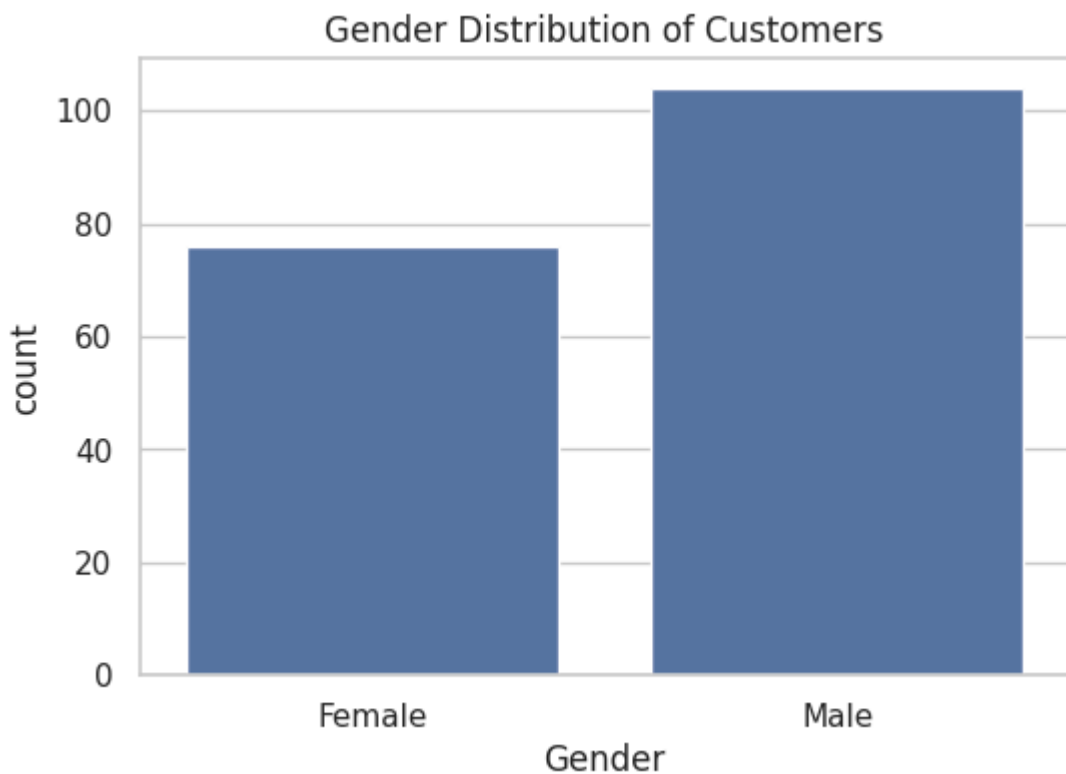
```
In [18]: df["Gender"].value_counts()
```

```
Out[18]: Gender
        Male    104
        Female   76
        Name: count, dtype: int64
```

```
In [19]: df["Gender"].value_counts(normalize=True) * 100
```

```
Out[19]: Gender
Male      57.777778
Female    42.222222
Name: proportion, dtype: float64
```

```
In [20]: plt.figure(figsize=(6,4))
sns.countplot(data=df, x="Gender")
plt.title("Gender Distribution of Customers")
plt.show()
```



Summary :

- ~58% of customers are male

Question : Does this distribution change by product?

```
In [21]: pd.crosstab(df["Product"], df["Gender"], normalize="index") * 100
```

```
Out[21]: Gender      Female      Male
Product
KP281    50.000000    50.000000
KP481    48.333333    51.666667
KP781    17.500000    82.500000
```

Summary:

- For KP281 and KP481, the gender distribution is close to even, with male and female customers represented in roughly equal proportions.
- For KP781, the distribution shifts with approximately 83% of purchases made by male customers, indicating a strong gender skew.

Marital Status Distribution

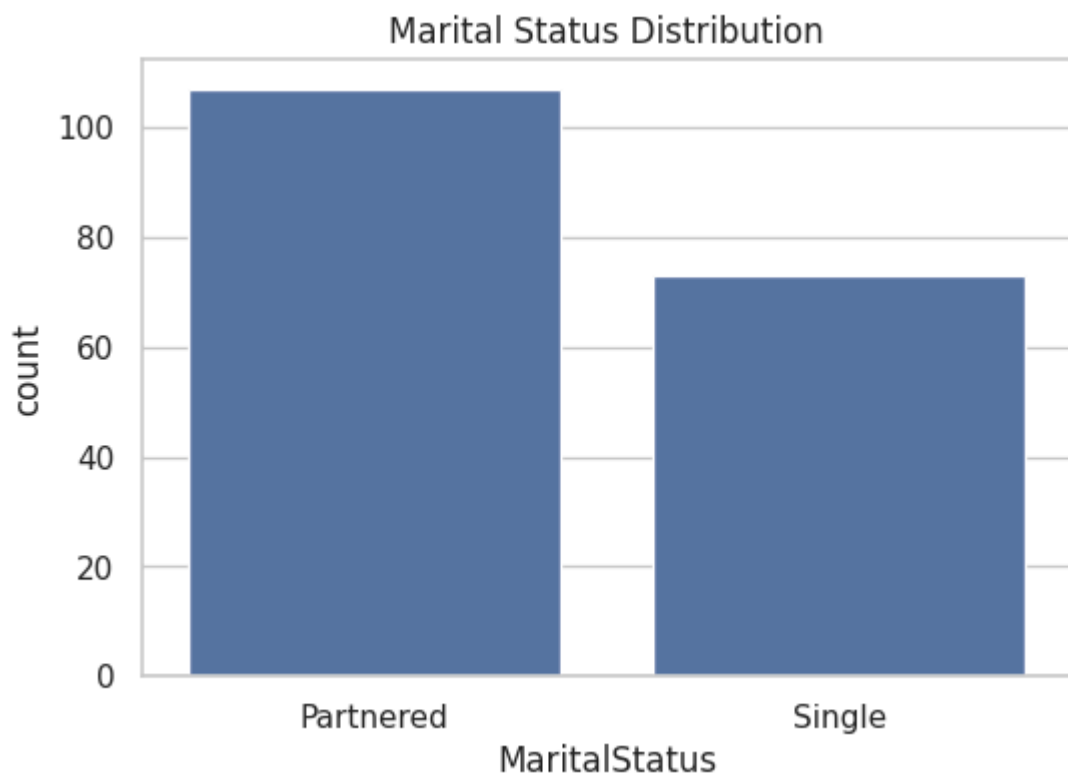
```
In [22]: df["MaritalStatus"].value_counts()
```

```
Out[22]: MaritalStatus  
Partnered    107  
Single        73  
Name: count, dtype: int64
```

```
In [23]: df["MaritalStatus"].value_counts(normalize=True) * 100
```

```
Out[23]: MaritalStatus  
Partnered    59.444444  
Single       40.555556  
Name: proportion, dtype: float64
```

```
In [24]: plt.figure(figsize=(6,4))  
sns.countplot(data=df, x="MaritalStatus")  
plt.title("Marital Status Distribution")  
plt.show()
```



```
In [25]: pd.crosstab(df["Product"], df["MaritalStatus"], normalize="index") * 100
```

Out [25]: **MaritalStatus** **Partnered** **Single**

Product		
KP281	60.0	40.0
KP481	60.0	40.0
KP781	57.5	42.5

Summary

- Across all three products, the marital status distribution is broadly similar. Partnered customers make up the majority for each product with ~60% and singles with 40%
- So, different products attract different genders, but roughly the same relationship statuses.

Fitness Distribution

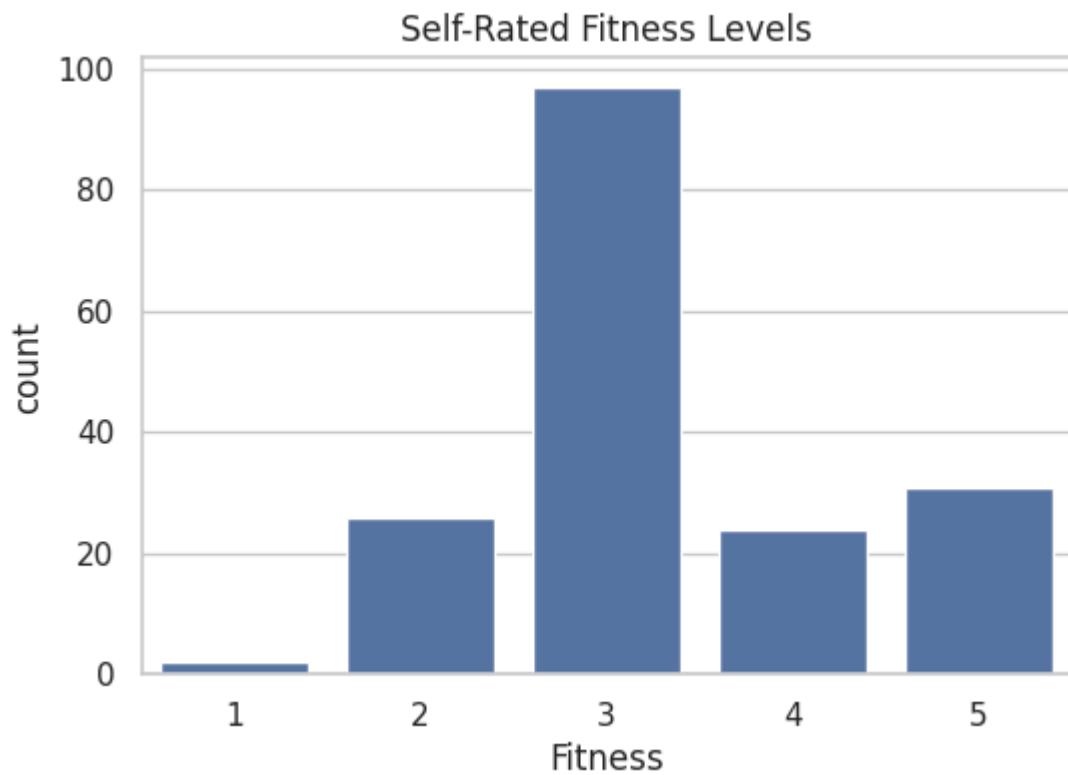
In [26]: `df["Fitness"].value_counts().sort_index()`

Out [26]: Fitness
 1 2
 2 26
 3 97
 4 24
 5 31
 Name: count, dtype: int64

In [27]: `df["Fitness"].value_counts(normalize=True).sort_index() * 100`

Out [27]: Fitness
 1 1.111111
 2 14.444444
 3 53.888889
 4 13.333333
 5 17.222222
 Name: proportion, dtype: float64

In [28]: `plt.figure(figsize=(6,4))
 sns.countplot(data=df, x="Fitness")
 plt.title("Self-Rated Fitness Levels")
 plt.show()`



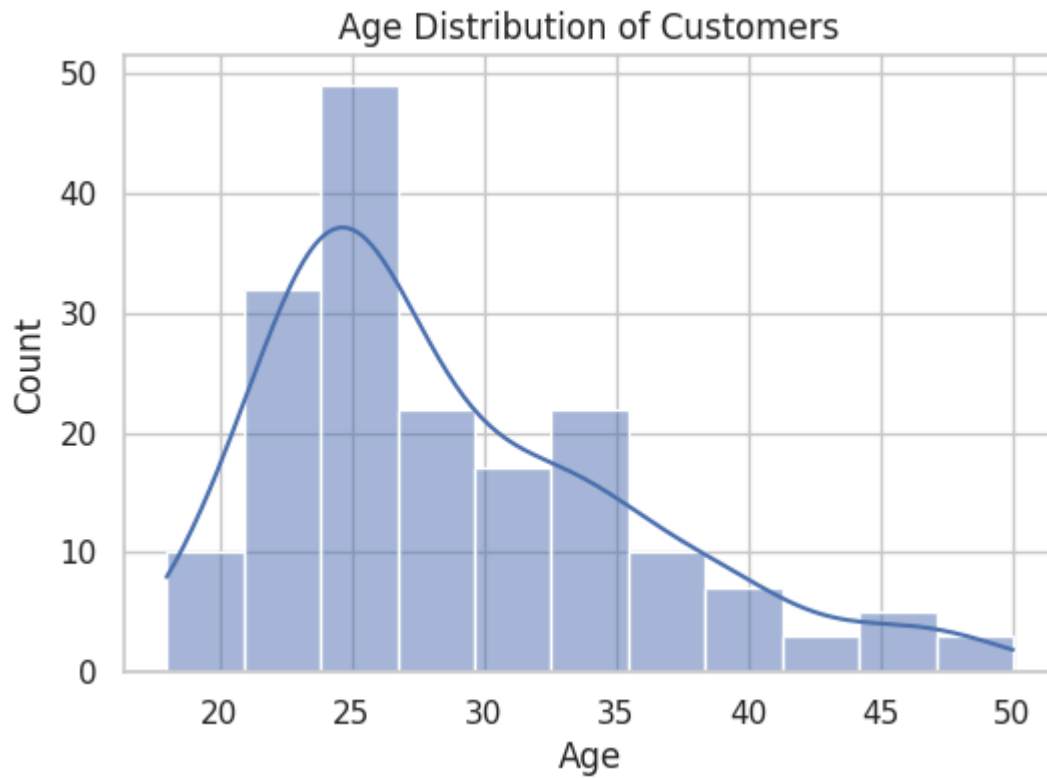
Summary :

- ~54%, More than half of the customers are at fitness level 3
- Only ~1% of users are at fitness level 1
- ~17% of users are at a fitness level of 5. Which is the highest level of fitness.

Analysis for Continuous Variables

Age Distribution

```
In [29]: plt.figure(figsize=(6,4))
sns.histplot(df["Age"], kde=True)
plt.title("Age Distribution of Customers")
plt.show()
```

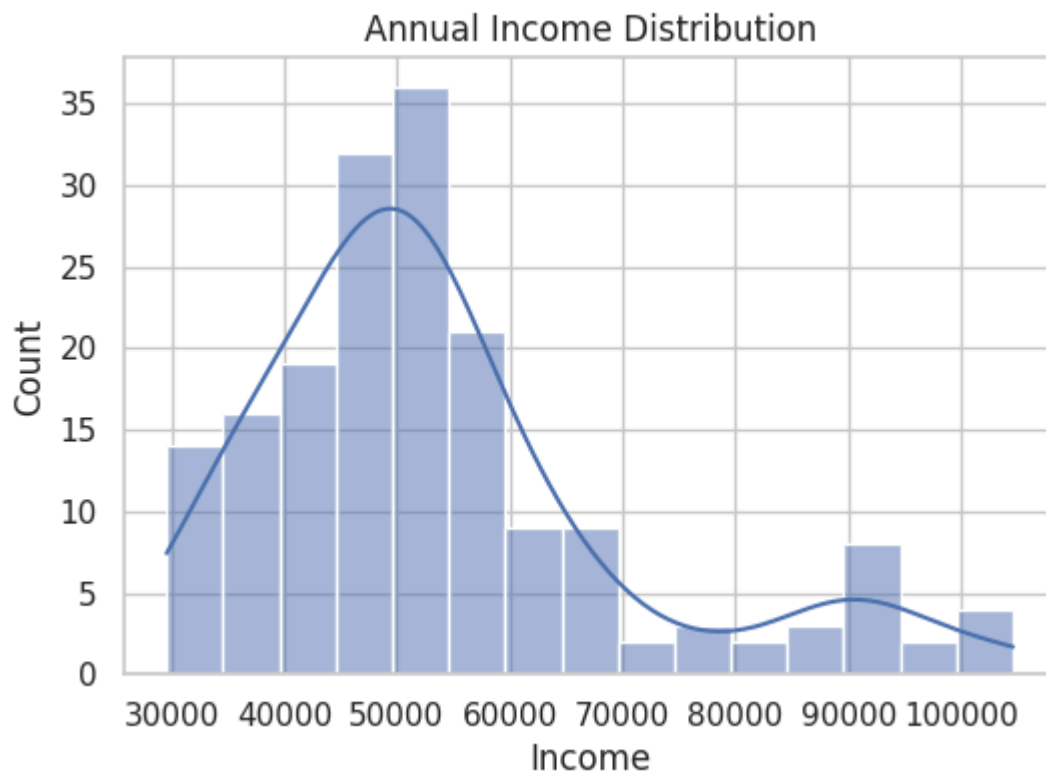


Summary :

- Age is right skewed and most of the users are in their 20s and early 30s
- And the density is highest around 25

Income Distribution

```
In [30]: plt.figure(figsize=(6,4))
sns.histplot(df["Income"], kde=True)
plt.title("Annual Income Distribution")
plt.show()
```

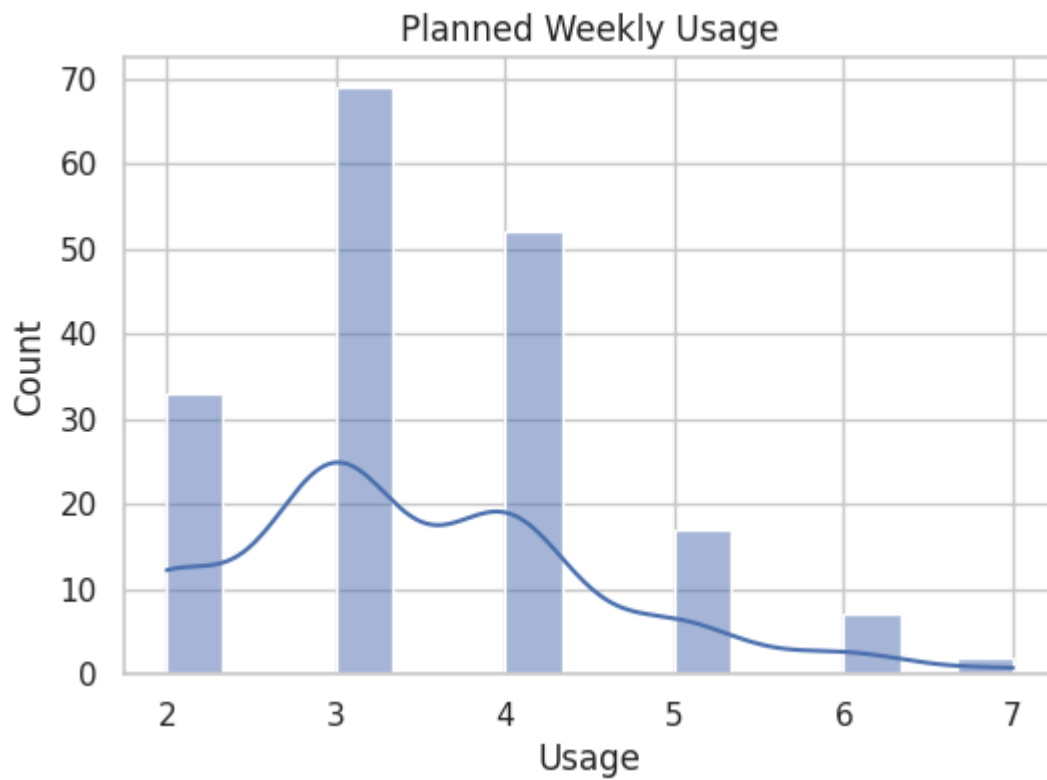


Summary

- Income is right skewed and majority is clustered around the 40,000 to 60,000 range.
- The peak of the distribution is around the 50,000 indicating the most common income level.
- A smaller number of customers earn significantly higher incomes, extending beyond 80,000 and up to 100,000+, forming a long right tail. These high-income individuals are relatively few but noticeably pull the average upward.
- The distribution suggests a predominantly middle-income customer base with a small, affluent segment at the upper end.

Usage distribution

```
In [31]: plt.figure(figsize=(6,4))
sns.histplot(df["Usage"], kde=True)
plt.title("Planned Weekly Usage")
plt.show()
```

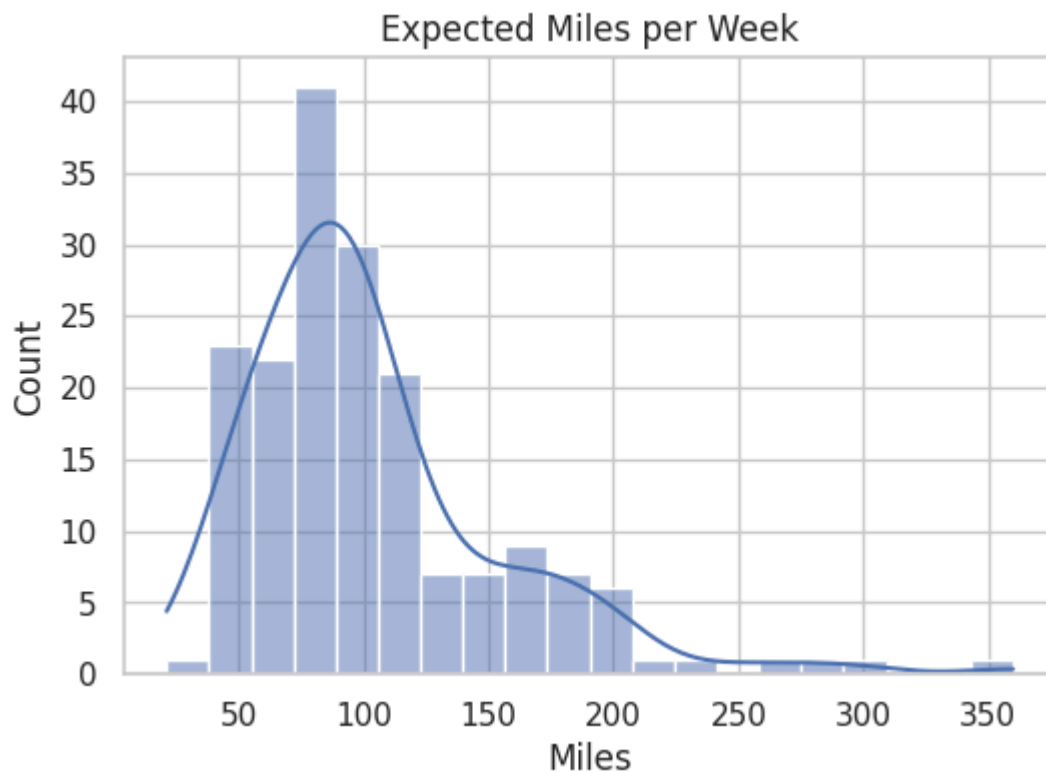


Summary

- The distribution is right skewed and highest around 3 and 4
- There is also a significant amount of density around 2
- After 4 the data is halved and then slowly fades.
- Which suggests that the majority of users are using the product 3-4 times each week.

Miles per week Distribution

```
In [32]: plt.figure(figsize=(6,4))
sns.histplot(df["Miles"], kde=True)
plt.title("Expected Miles per Week")
plt.show()
```

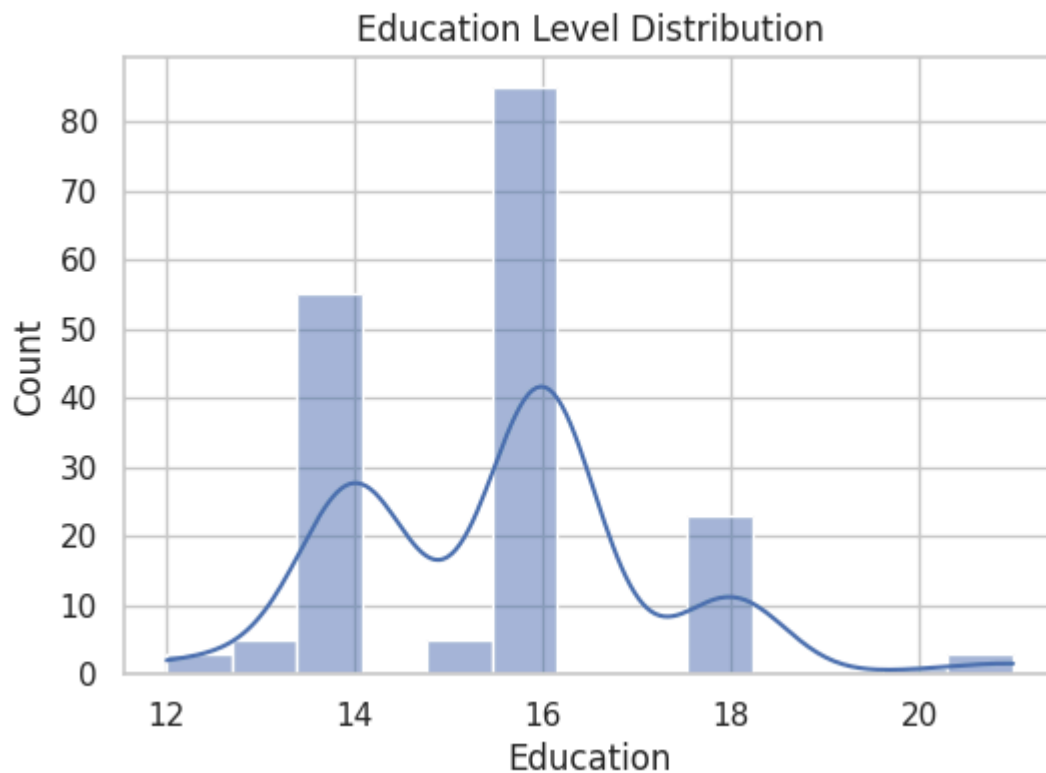


Summary :

- Most weeks cluster between about 50 and 120 miles, with a clear peak around 80–100 miles, indicating that this is the typical weekly mileage. The distribution is right-skewed, meaning a small number of weeks involve much higher mileage. These rare weeks extend the tail of the distribution past 200 miles, reaching over 300 miles. As a result, high-mileage weeks are uncommon but have a noticeable impact on the overall distribution.

Education distrubution

```
In [33]: plt.figure(figsize=(6,4))
sns.histplot(df["Education"], kde=True)
plt.title("Education Level Distribution")
plt.show()
```



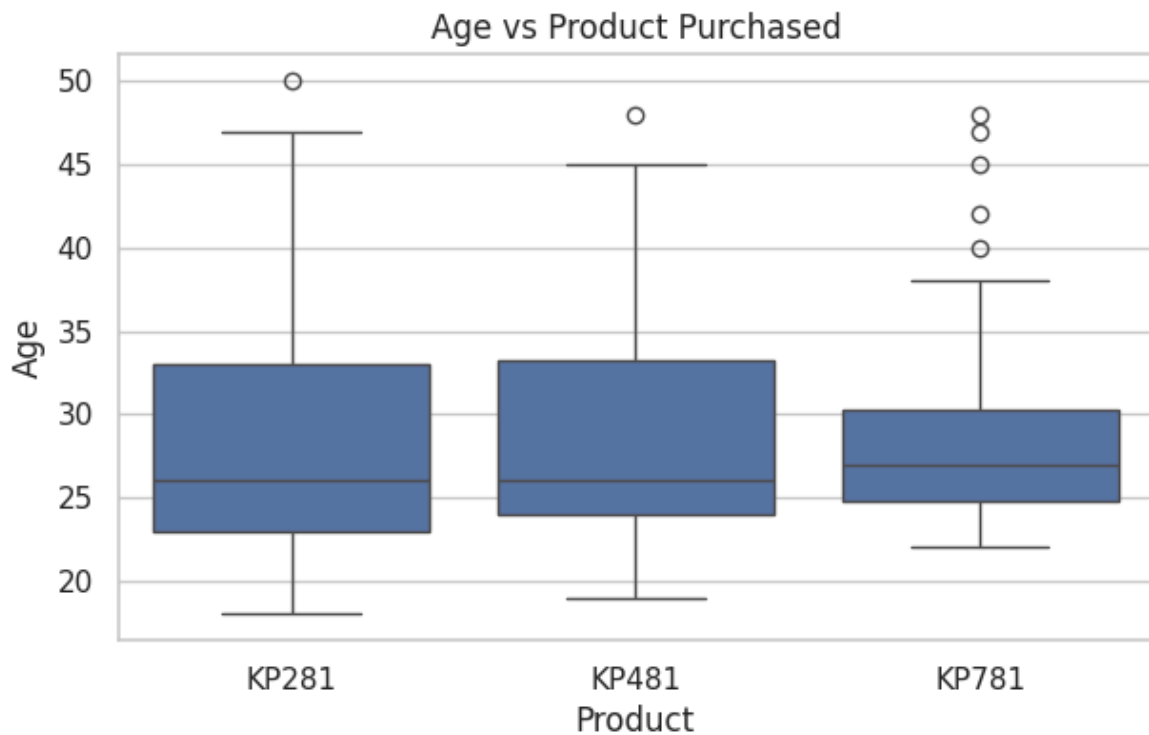
Summary

The plot shows the distribution of education levels measured in years. Most individuals fall between 14 and 16 years of education, with 16 years (roughly a bachelor's degree) being the most common. A smaller group appears around 14 years (some college or an associate level), and fewer individuals have higher education levels around 18–21 years (graduate or professional degrees). Very few are at the lower end near 12 years (high school). Overall, the distribution is centered on undergraduate-level education with a thinner tail toward advanced degrees.

Bivariate Visual Analysis

Age VS Product

```
In [34]: plt.figure(figsize=(7,4))
sns.boxplot(data=df, x="Product", y="Age")
plt.title("Age vs Product Purchased")
plt.show()
```

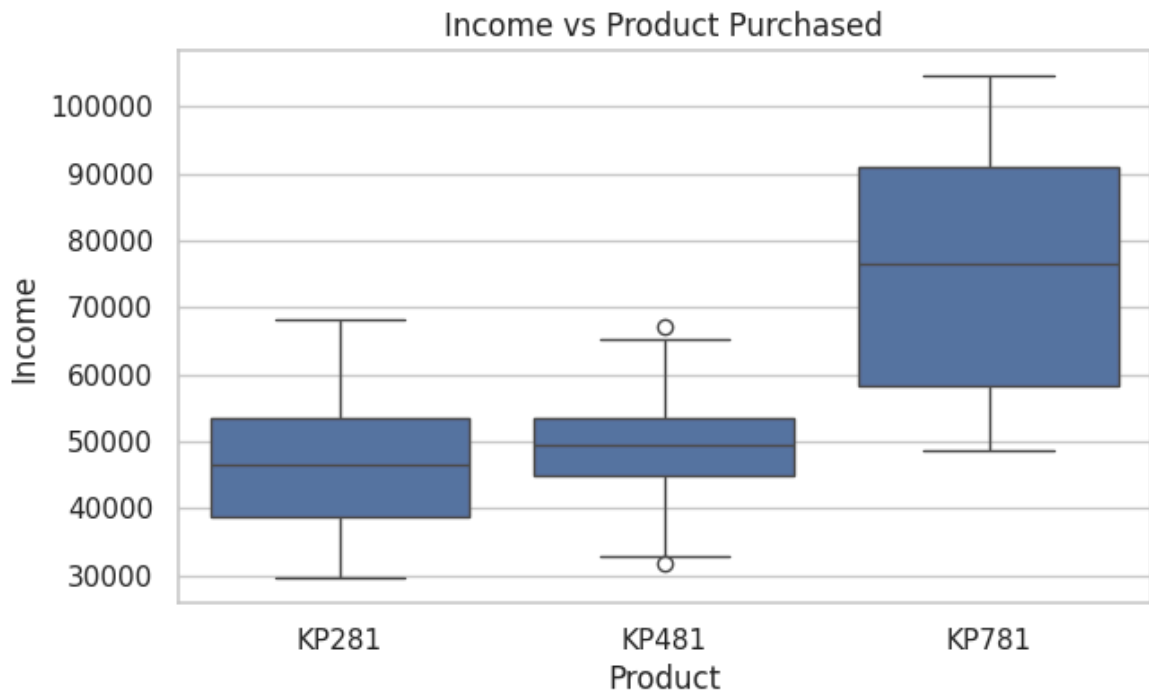


Summary

For all three products, the median customer age is in the mid-20s, showing that the core customer base is similar across products. KP281 and KP481 have wider age spreads, indicating a more diverse age range of buyers, including some older customers. KP781 shows a slightly tighter distribution, suggesting it appeals more consistently to a narrower age group. All products have a few older outliers, but overall, age differences between product purchasers are modest.

Income VS Product

```
In [35]: plt.figure(figsize=(7,4))
sns.boxplot(data=df, x="Product", y="Income")
plt.title("Income vs Product Purchased")
plt.show()
```

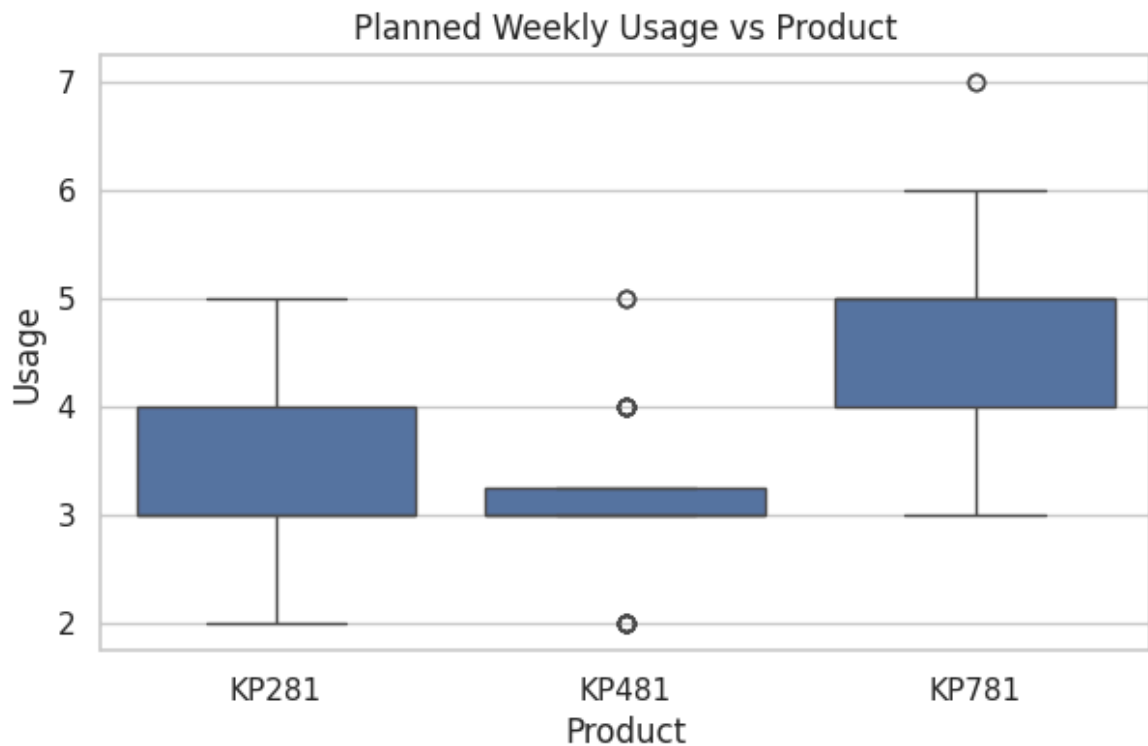


Summary

For the product KP281 and KP481 the income level is centered around mid 40,000 to mid 50,000 range. In contrast, KP781 is purchased by significantly high income customers with a median of mid 70,000 and a much wider spread extending 100,000. This indicated KP781 appeals to more affluent segment while KP281 and KP481 target middle-income customers.

Usage (time) Vs Products

```
In [36]: plt.figure(figsize=(7,4))
sns.boxplot(data=df, x="Product", y="Usage")
plt.title("Planned Weekly Usage vs Product")
plt.show()
```



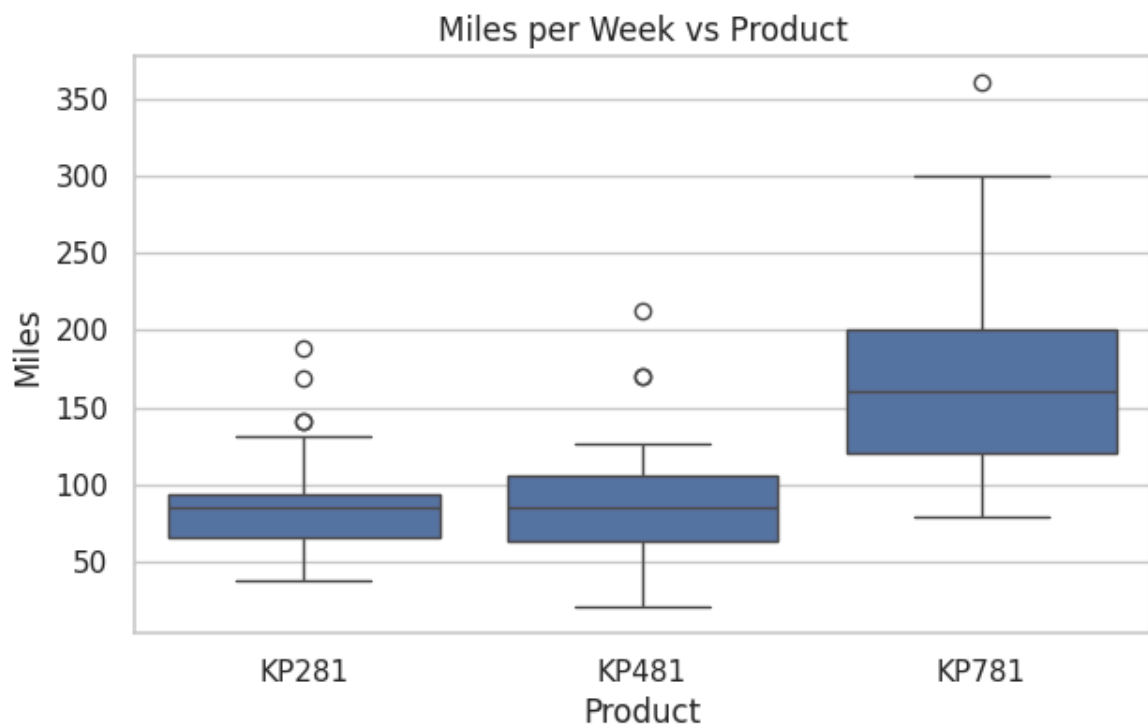
Summary

There is a tight spread for KP481 which is focused on 3 planned usage per week with a few outliers. While the KP281 and KP781 are more spread out around the 3-4 and 4-5 range respectively with a extension of 1 on each side fo the spread.

KP481 is most consistent and lowest used around 3 times per week and KP781 is the most used with 4-5 times per week.

Mile per week s Product

```
In [37]: plt.figure(figsize=(7,4))
sns.boxplot(data=df, x="Product", y="Miles")
plt.title("Miles per Week vs Product")
plt.show()
```

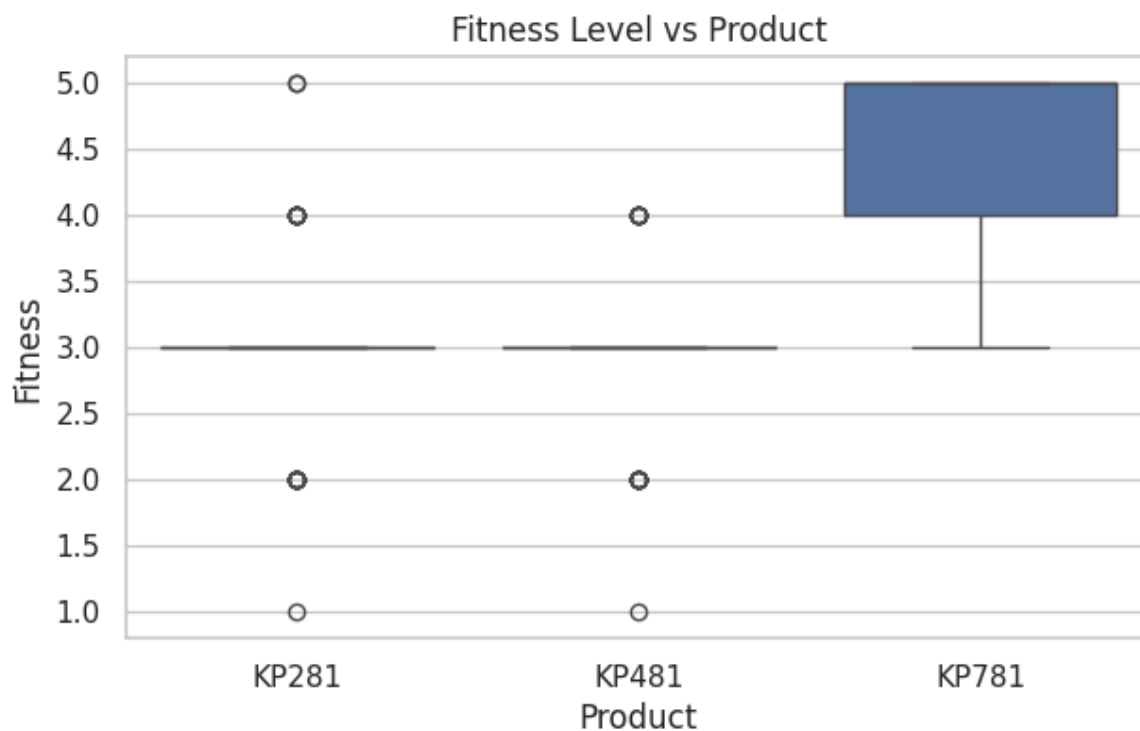


Summary

KP281 and KP481 are centered around 50-100 with a tight spread while the KP781 is more widely spread with a center around 150 and a few outliers extending till 350. Clearly stating that KP781 is used by more experienced customers with high level of activity.

Fitness Vs Products

```
In [38]: plt.figure(figsize=(7,4))
sns.boxplot(data=df, x="Product", y="Fitness")
plt.title("Fitness Level vs Product")
plt.show()
```

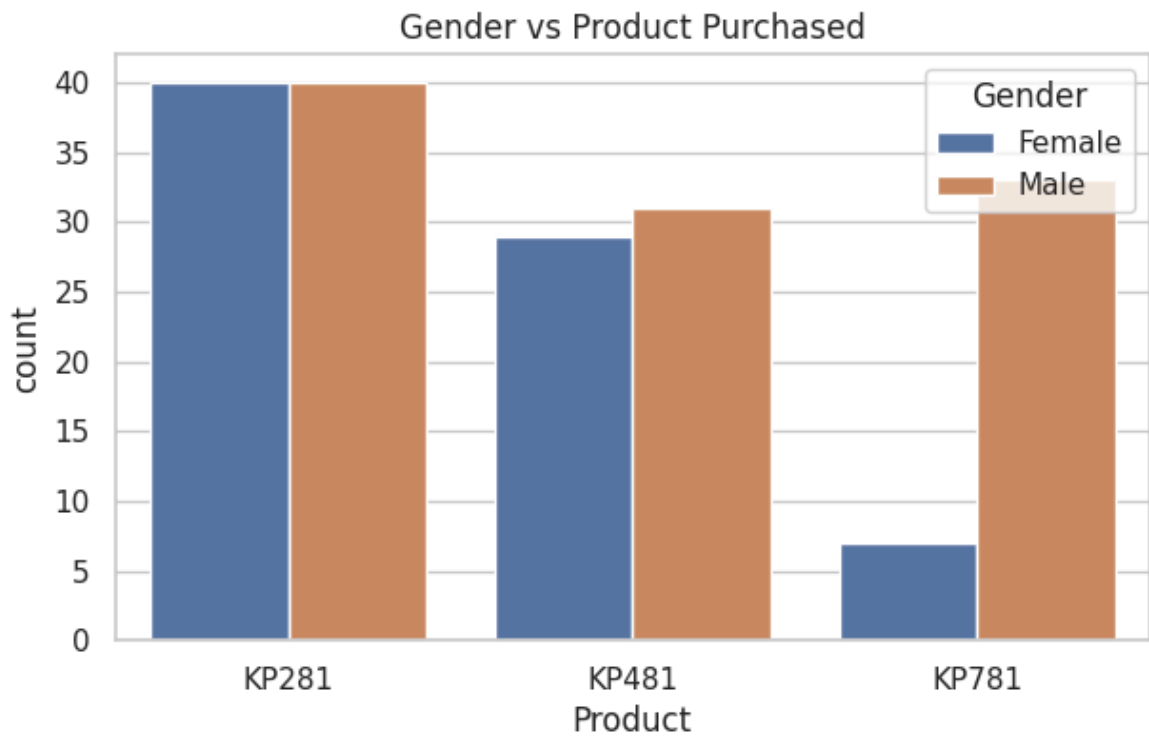


Summary

KP281 and KP481 are centered at 3 fitness level with a few outliers while the KP781 is centered around the range 4-5 with a more wider spread suggesting that the KP781 is used by more experienced customers.

Gender Vs Product

```
In [39]: plt.figure(figsize=(7,4))
sns.countplot(data=df, x="Product", hue="Gender")
plt.title("Gender vs Product Purchased")
plt.show()
```

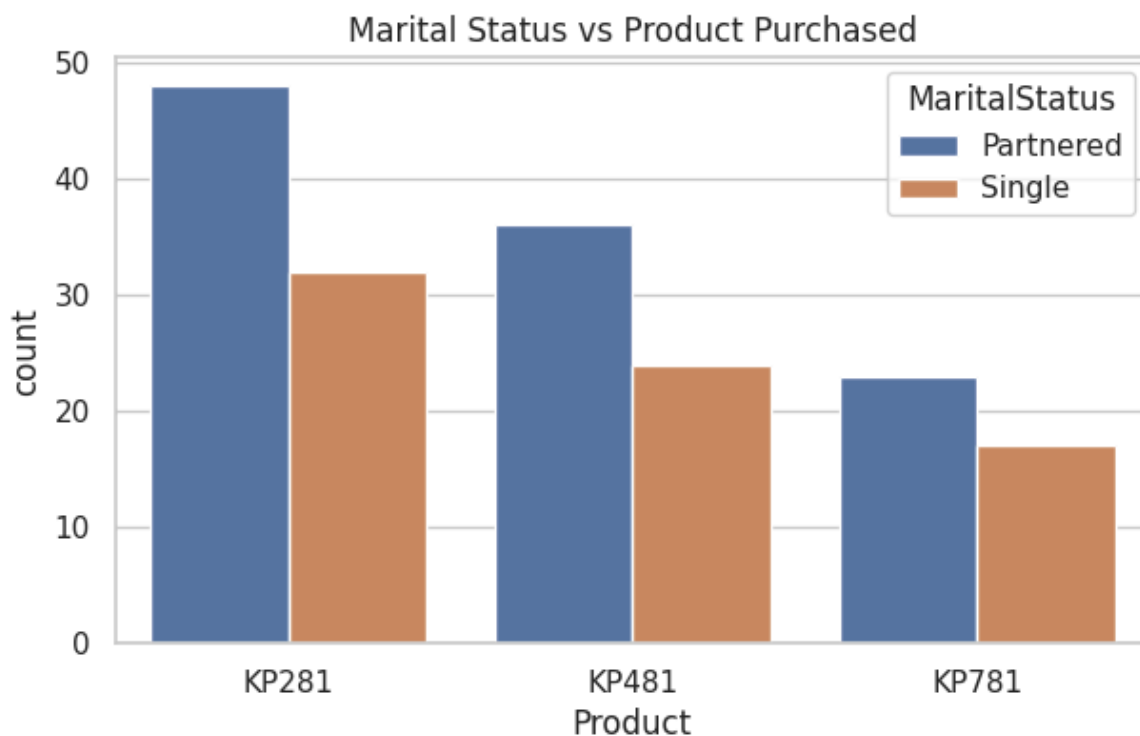


Summary

For KP281 and KP481 the difference between the gender is negligible but KP781 had a huge imbalance in ratio with a greater than 3 times male population over female. Overall, gender differences become more pronounced for the higher-end product, suggesting KP781 appeals much more strongly to male customers.

Marital Status Vs Product

```
In [40]: plt.figure(figsize=(7,4))
sns.countplot(data=df, x="Product", hue="MaritalStatus")
plt.title("Marital Status vs Product Purchased")
plt.show()
```



Summary

The chart shows that partnered customers consistently outnumber single customers across all three products. The difference is largest for KP281, where partnered buyers are much more common than singles. As we move to KP481 and then KP781, the overall number of customers declines and the gap between partnered and single customers narrows, though partnered customers still remain the majority for each product. This suggests that while all products appeal more to partnered individuals, that preference is strongest for KP281 and less pronounced for the higher-tier products.

Final Product Summary

KP281 is designed for beginners and casual users. It is mostly purchased by customers with lower to mid income, moderate fitness levels, and lower planned usage. This product appeals to partnered first-time treadmill buyers who want an affordable option for regular walking or light running.

KP481 targets moderately active users who plan to use the treadmill more frequently. Customers buying this product generally have higher income and fitness levels than KP281 buyers. It serves as a balanced option for users who want better features without moving into the premium price range.

KP781 is purchased by highly active and fitness-focused customers. Buyers typically have higher income, higher fitness ratings, and expect to run more miles per week. This product caters to serious runners and health enthusiasts looking for advanced features and long-term performance.

Customer Personas & Business recommendations :

Persona 1: The Beginner Walker (KP281)

- Age: Young to middle-aged adults
- Income: Low to mid-range
- Fitness Level: 1–3
- Usage: 2–3 times per week
- Goal: Start a fitness routine, stay healthy, lose weight

Mindset: "I want to get fitter, but I don't need anything fancy. It should be affordable and easy to use."

Recommendations :

- Position KP281 as a starter treadmill.
- Emphasize ease of use, affordability, and health benefits.

Actions :

- Target first-time buyers through beginner fitness campaigns.
- Bundle with basic fitness plans or walking programs.
- Promote KP281 in stores and online as "perfect for getting started".

Persona 2: The Consistent Exerciser (KP481)

- **Age:** Middle-aged adults
- **Income:** Mid-range
- **Fitness Level:** 3–4
- **Usage:** 3–5 times per week
- **Goal:** Maintain fitness, jog regularly, balance work and health

Mindset:

"I work out regularly and want a treadmill that can keep up with me without being too expensive or overcomplicated."

Recommendations:

- Position KP481 as a reliable mid-level treadmill for regular users.
- Highlight the balance between performance, comfort, and price.

Actions:

- Upsell KP481 to existing KP281 customers looking to upgrade.
- Target working professionals through digital and corporate wellness campaigns.
- Promote durability and improved workout features over entry-level models.

Persona 3: The Serious Runner (KP781)

- **Age:** Middle-aged to older adults

- **Income:** High
- **Fitness Level:** 4–5
- **Usage:** 5–6 times per week
- **Goal:** Performance improvement, endurance training, long-term fitness

Mindset:

"Fitness is a priority for me. I want advanced features, high performance, and equipment that supports intense training."

Recommendations:

- Position KP781 as a premium, high-performance treadmill.
- Emphasize advanced features, durability, and long-term value.

Actions:

- Target high-income and high-fitness customers with personalized marketing.
- Highlight performance metrics, training programs, and premium build quality.
- Offer add-ons such as extended warranty or expert-led fitness content.

In []: